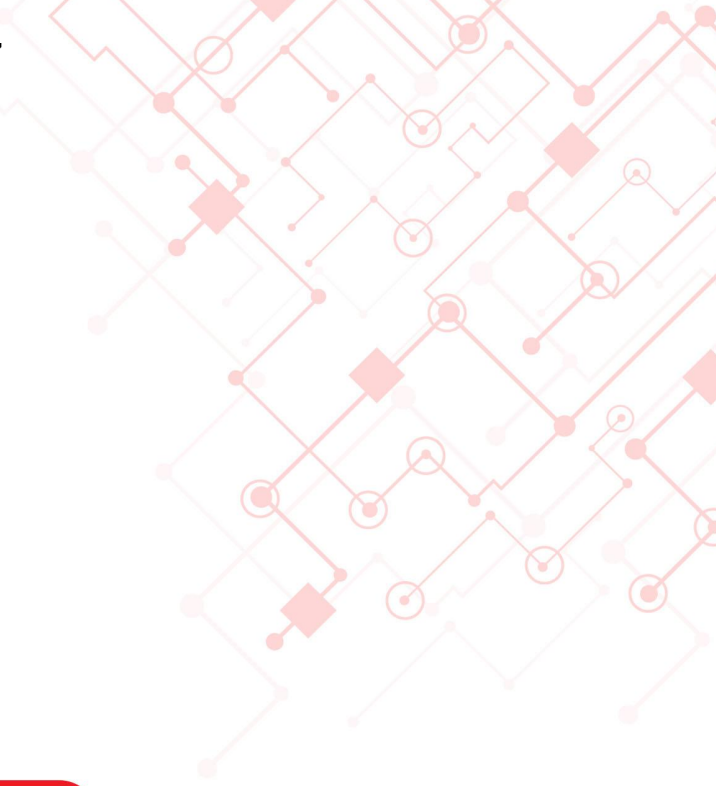




ATP

WAVES



- 1 A student walks away from a street lamp at night and notices that his shadow changes.

The student investigates shadows in a darkened laboratory. He uses a lamp on a stand and a vertical card, as shown in Fig. 1.1.

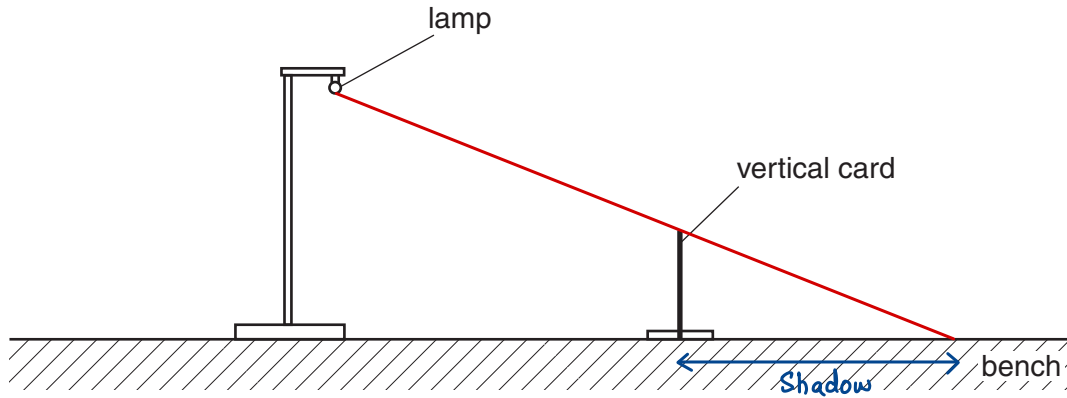


Fig. 1.1

- (a) (i) On Fig. 1.1, draw a ray of light from the lamp to the bench, just passing over the top of the card. [1]
- (ii) The region of the bench that does not receive light from the lamp is the 'shadow' of the card.
On Fig. 1.1, draw a line on the surface of the bench to indicate this shadow. [1]

- (b) The card is moved along the bench away from the lamp.

State what happens to the shadow of the card.

Shadow will get less dark.

.....[1]

- (c) The shadow of the card produced in the laboratory is much darker than the shadow of the student in the street.

Suggest a reason for this.

Not all the light of street lamp is reaching student & moonlight is also present.

.....[1]

- 2 A student investigates the effect of a converging lens on light from the Sun.

The student uses a converging lens to produce a clear image of the Sun on a piece of white card, as shown in Fig. 2.1.

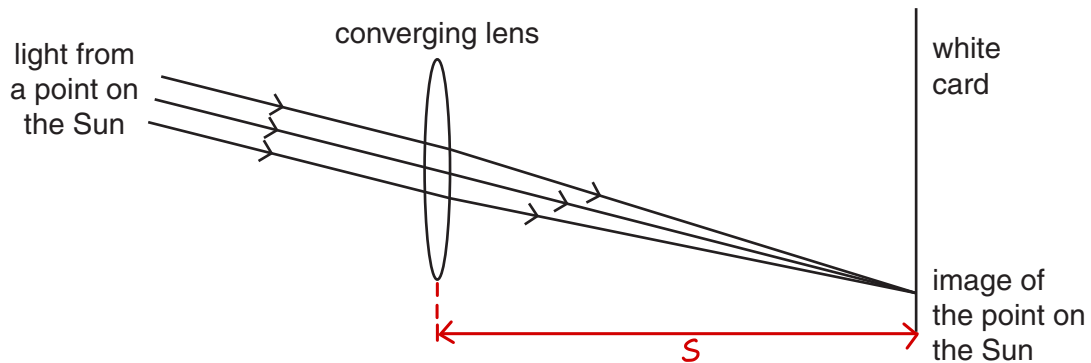


Fig. 2.1 (not to scale)

The student measures the perpendicular distance from the centre of the lens to the white card.

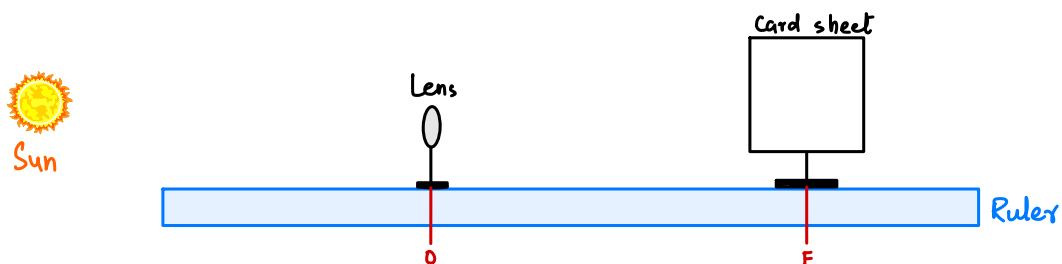
- (a) (i) On Fig. 2.1, mark this distance and label it s . [1]

- (ii) State the name given to this distance.

Focal Length

[1]

- (b) Describe in detail the experimental techniques used to obtain an accurate value for this distance.



Arrange the apparatus as shown, both lens & cardsheet should be placed near the ruler. Move the lens until clear image of sun appears on white sheet. Measure distance b/w F & O. This is focal length

[3]

- 3 A student investigates the refraction of light through a parallel-sided glass block.

Fig. 3.1 shows the student's diagram. A line represents a ray of light arriving at the upper surface of the block.

The student forgets to mark the lower surface of the block.

The student draws two crosses M_1 and M_2 on the path of the emergent ray.

- (a) (i) On Fig. 3.1, draw the normal through the point where the ray enters the block. [1]
 (ii) Mark the angle of incidence and label it i . [1]
 (iii) Measure the angle of incidence i .

$$i = \dots\dots\dots 30^\circ \dots\dots\dots [1]$$

- (b) The distance between the upper surface and the lower surface of the block is 5.0 cm.

On Fig. 3.1,

- (i) draw the lower surface of the block, [1]
 (ii) complete the path of the emergent ray from the lower surface of the block through M_1 and M_2 , [1]
 (iii) complete the path of the ray within the block. [1]
- (c) Measure the angle of refraction r inside the block.

$$r = \dots\dots\dots 16^\circ \dots\dots\dots [1]$$

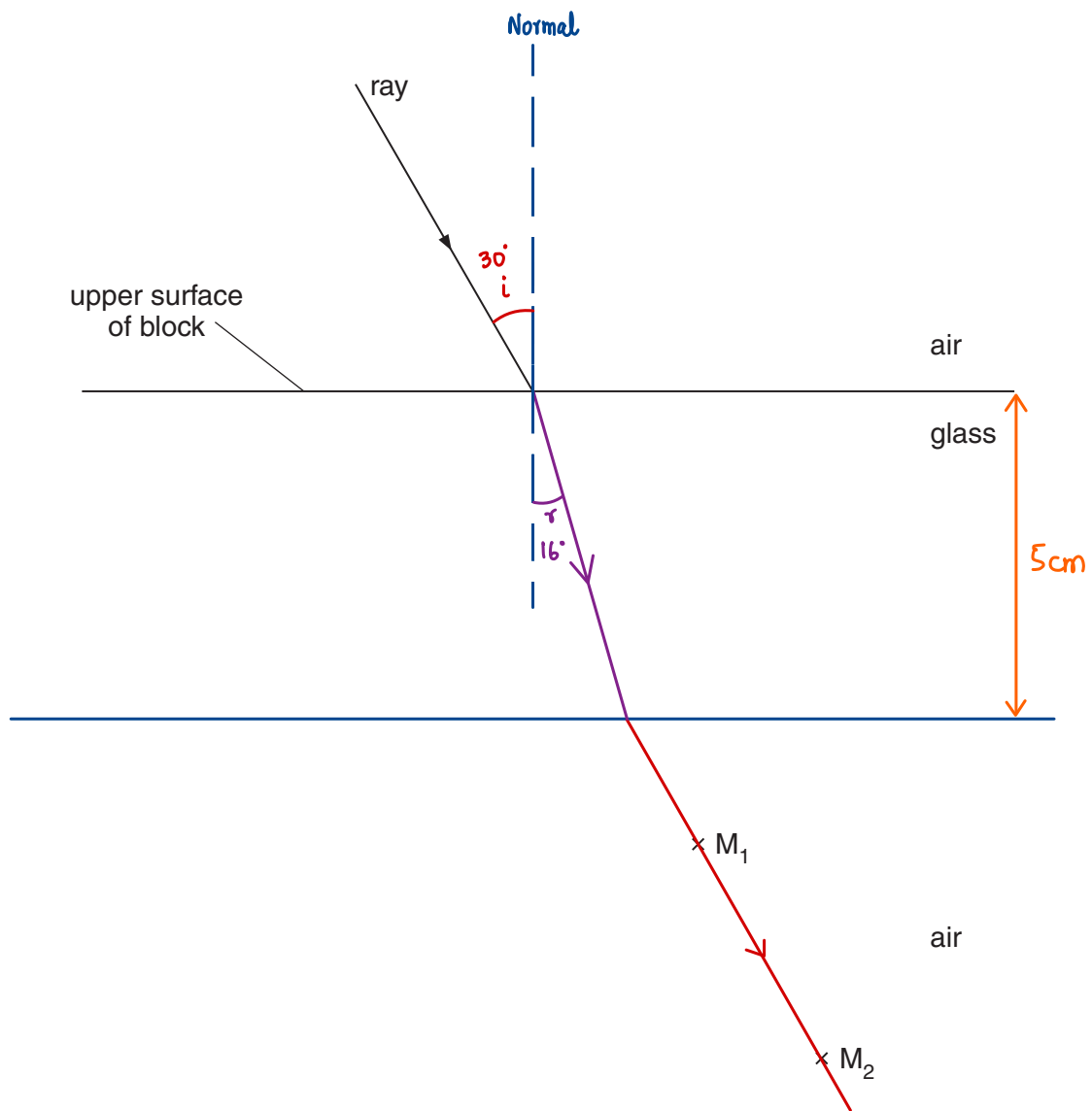


Fig. 3.1 (full size)

- 4 A student measures the focal length of a converging lens.

The student sets up the apparatus as shown in Fig. 4.1.

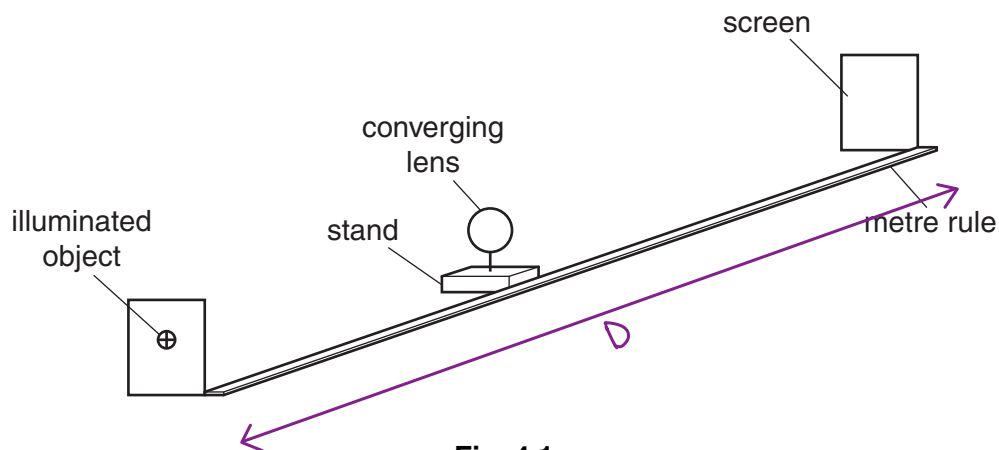


Fig. 4.1

- (a) The distance D between the illuminated object and the screen is fixed initially at 1.00 m.
- The image on the screen is blurred. Describe the adjustment that the student makes to obtain a sharply focused image, without changing D .
Move the lens towards or away from screen until sharpest image is formed. [1]
 - The focused image is at the top of the screen. Describe one adjustment that the student makes to move the focused image to the centre of the screen.
Move the screen upwards such that center of object, lens & screen are aligned. [1]
 - The lens is in a holder on a stand. The position of the centre of the lens is to be marked accurately on the stand. Describe how the student does this.
By avoiding parallax error, mark a dot parallel to the pin holding the lens. [1]

- (b) The student finds that, for the same value of D , there are two positions of the lens that produce a focused image on the screen. These positions are at distances u_1 and u_2 from the illuminated object, as shown in Fig. 4.2.

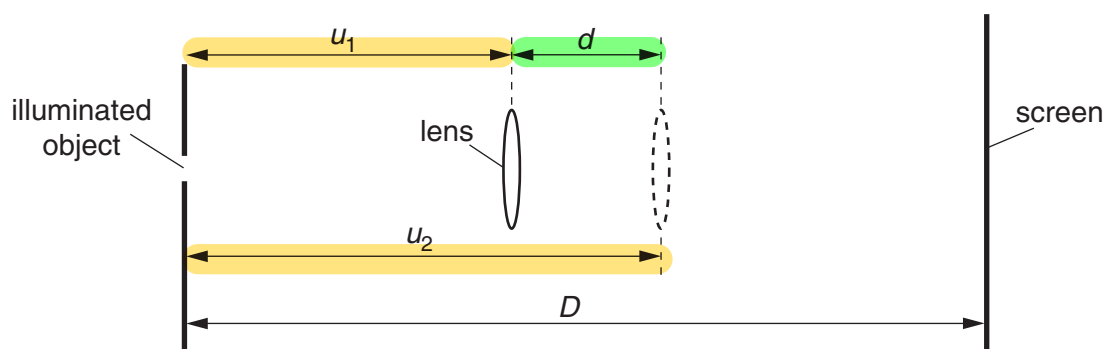


Fig. 4.2

The two distances u_1 and u_2 are measured for $D = 1.00\text{ m}$.
The values obtained are $u_1 = 0.42\text{ m}$ and $u_2 = 0.56\text{ m}$.

- (i) Calculate the distance d between the two lens positions.

$$0.56 - 0.42$$

$$d = 0.14\text{ m} \dots\dots\dots [1]$$

- (ii) Theory shows that the focal length f of the lens is given by the relationship

$$f = \frac{(D^2 - d^2)}{4D} = \frac{(1^2 - 0.14^2)}{4 \times 1}$$

Calculate f .

$$f = 0.245\text{ m} \dots\dots\dots [1]$$

- (c) The student repeats the experiment for different values of D and calculates values of $(D^2 - d^2)$ each time.

Fig. 4.3 shows the student's results.

D/m	$(D^2 - d^2)/\text{m}^2$
1.10	1.08
1.20	1.18
1.30	1.29
1.40	1.38
1.50	1.47

Fig. 4.3

- (i) On Fig. 4.4, plot the graph of $(D^2 - d^2)/m^2$ on the y-axis against D/m on the x-axis. Start your graph from $(D^2 - d^2) = 0.90 \text{ m}^2$ and $D = 1.00 \text{ m}$. Draw the straight line of best fit.

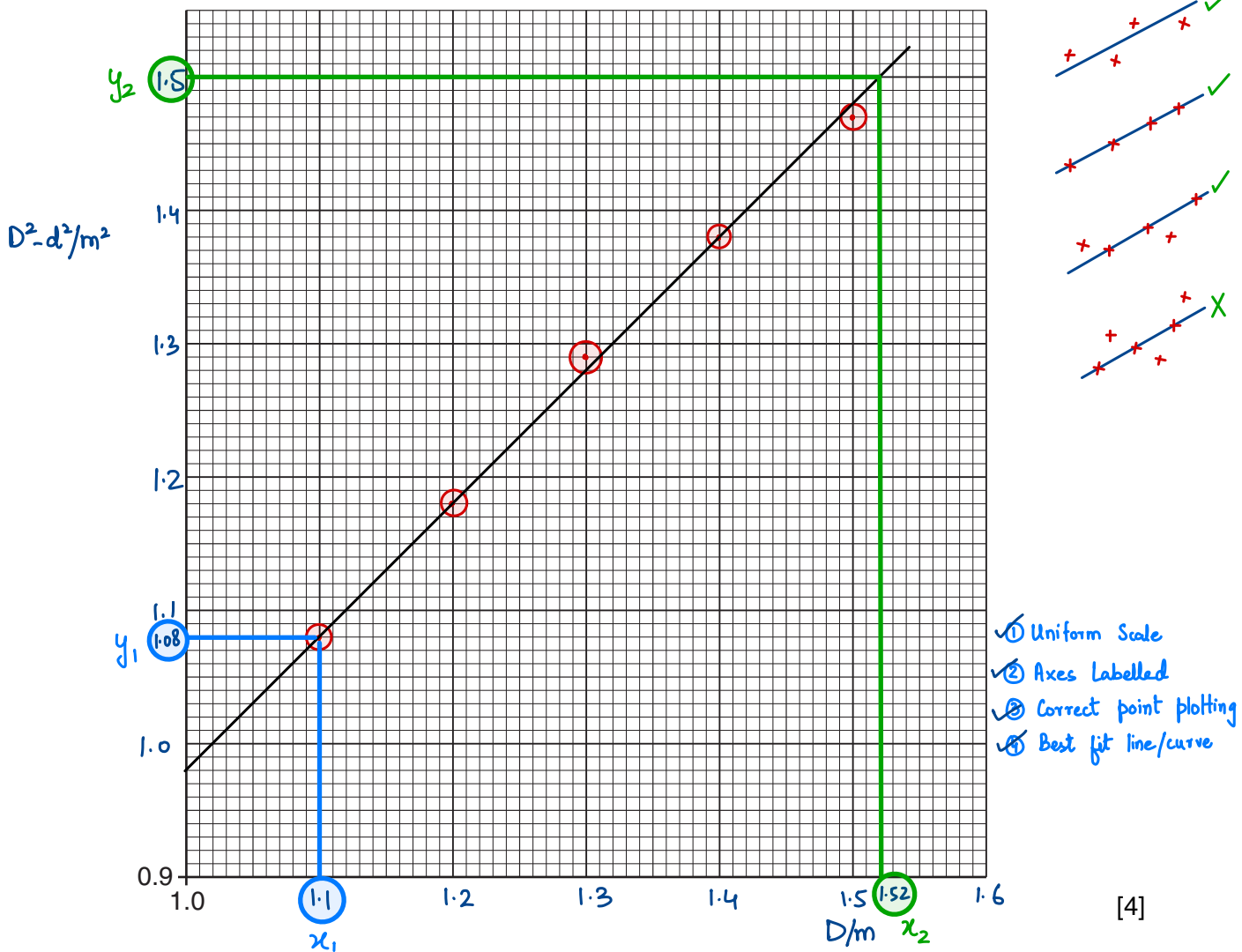


Fig. 4.4

- (ii) Determine the gradient of the line of best fit. Show your working clearly.

$$\text{grad} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1.5 - 1.08}{1.52 - 1.1} \frac{\text{m}^2}{\text{m}}$$

$$\text{gradient} = 1.0 \text{ m} \quad [2]$$

- (iii) Calculate f using the relationship

$$f = \frac{\text{gradient}}{4} = \frac{1}{4}$$

$$f = 0.25 \text{ m} \quad [1]$$

- (d) Explain why it is better to determine f using the method in (c) rather than the method in (b).

Method in (c) is more accurate as it is giving an averaged result of the experiment. [1]

- 5 A semi-circular glass block is used to measure the critical angle for light in glass.

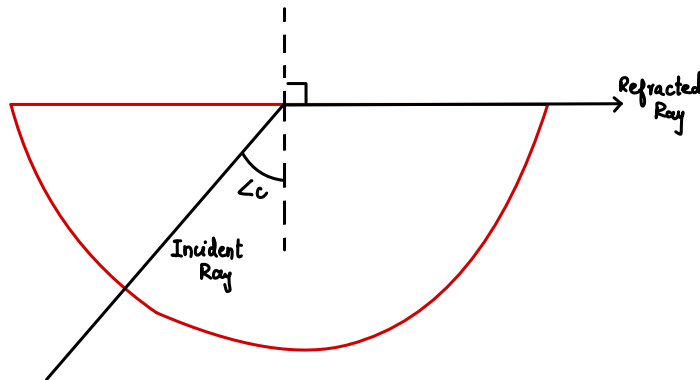
Fig. 5.1 on page 11 shows the semi-circular block on a sheet of white paper.

A student shines a narrow ray of light from a ray-box through the semi-circular block and draws two crosses M_1 and M_2 on the emergent ray.

- (a) (i) On Fig. 5.1, complete the path of the ray after it emerges from the block. [1]
 (ii) On Fig. 5.1, draw the path of the ray through the block. [1]
 (iii) Using Fig. 5.1, measure the angle of incidence i where the light passes from glass to air.

$i = 28^\circ$ [1]

- (b) Describe how the student adjusts the apparatus to measure the critical angle for glass.
 You may use a diagram to help your explanation.



Increase angle of incidence so that angle of refraction is 90° .
 This angle of incidence from denser to less dense is critical angle.

..... [2]

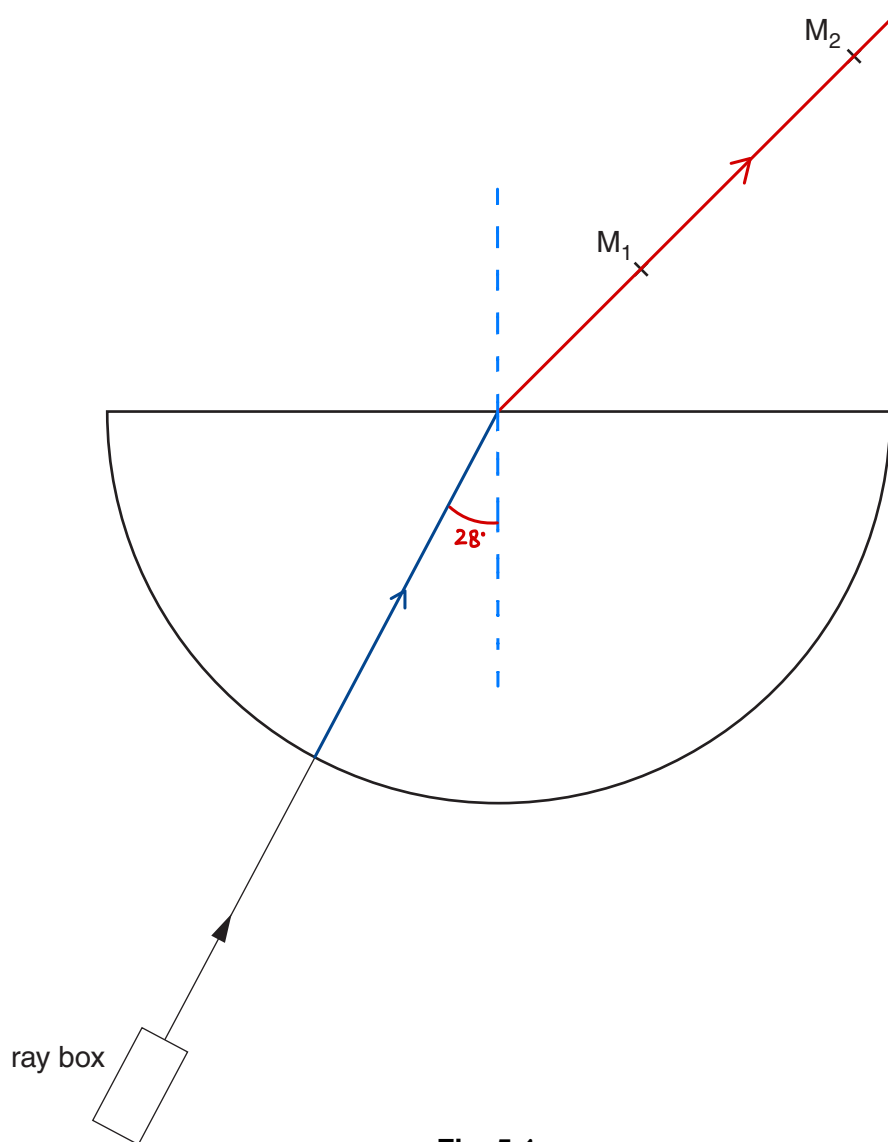


Fig. 5.1

- (c) Fig. 5.2 shows the incident ray used by another student. The ray in the block strikes the glass-air surface at the critical angle.

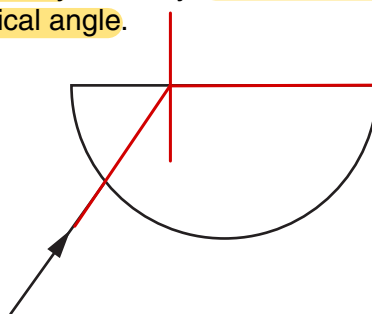


Fig. 5.2

Explain why it is more difficult to measure the critical angle with this arrangement than with the arrangement in (b).

Difficult to draw normal as ray will not pass through center of glass block.

[1]

Home Work.

- 6 A teacher demonstrates a property of sound.

The teacher hangs an electric bell inside a glass bell-jar, as shown in Fig. 6.1.

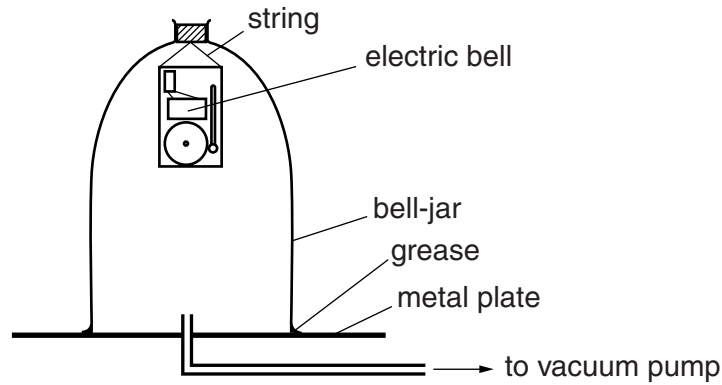


Fig. 6.1

The bell is ringing when the bell-jar is placed on the metal plate. The air can then be removed from the bell-jar using a vacuum pump.

- (a) Explain why the bell-jar must have thick walls.

.....
 [1]

- (b) Explain why the teacher puts a layer of grease between the bell-jar and the metal plate.

.....
 [1]

- (c) The students can see the bell vibrating and can hear it ringing.

- (i) Describe and explain what the students hear as the air is removed.

.....

 [2]

- (ii) When the air is removed, the students can still see the bell vibrating.

State what this experiment demonstrates about light.

..... [1]

- (d) Explain why the teacher hangs the bell from string inside the bell-jar rather than standing the bell on the metal plate.

.....
 [1]

- 7 A student investigates the speed of water waves.

A rectangular plastic tray has a layer of water in the bottom, as shown in Fig. 7.1.

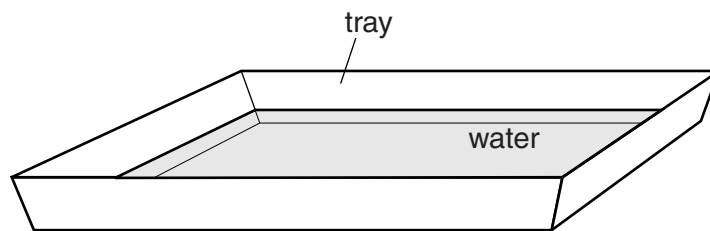


Fig. 7.1

One end of the tray is lifted gently a small distance, as shown in Fig. 7.2a. When the water is still, the tray is lowered quickly. This causes a wave to travel across the surface of the water, as shown in Fig. 7.2b.

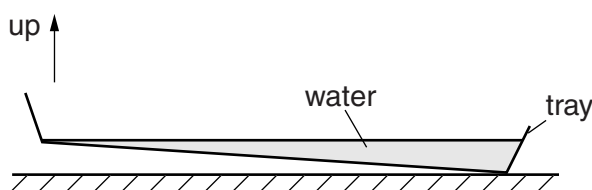


Fig. 7.2a

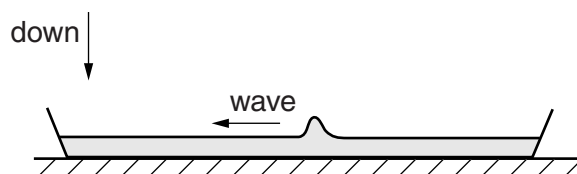


Fig. 7.2b

The wave is reflected by the ends of the tray and travels backwards and forwards several times.

- (a) The student estimates the time for the wave to travel one length of the tray to be about one second.

Suggest a reason why the student decides to measure the time for the wave to travel five lengths of the tray.

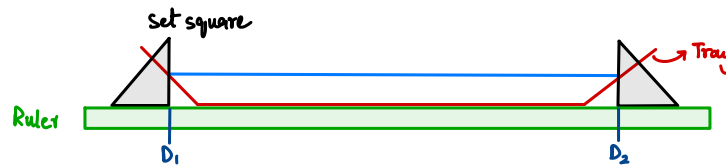
To minimize human reaction error occurred during time measurement.
 & then take an average. [1]

- (b) The ends of the tray slope outwards.

- (i) Explain why this makes it difficult to measure accurately the distance travelled by the wave.

Difficult to place the ruler as length of tray varies along the slope. [1]

- (ii) Describe a possible method of measuring the distance travelled by the wave. You may use a diagram in your answer.



Set the apparatus as shown & measure the length b/w D_1 & D_2 using set squares & meter rule. [1]

- (c) The student measures the depth d of the water. He has two identical 30 cm rulers. Part of one ruler is shown in Fig. 7.3.

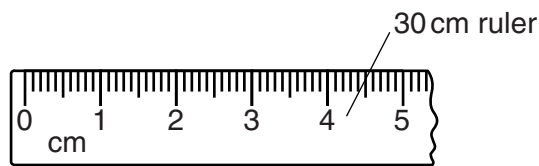
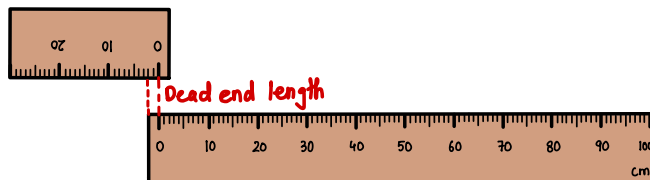


Fig. 7.3

- (i) Explain why it is difficult to measure d using only one of these rulers.

Dead end length is also adding up in the depth. [1]

- (ii) Describe a method of measuring d using the two rulers. You may use a diagram in your answer.



Use 2nd ruler to measure dead end length of 1st ruler then add dead end length in the depth [2]

- (d) The student repeats the experiment for different values of d and calculates the speed v of the wave for each value of d .

Fig. 7.4 shows the student's results.

d/cm	$v/(\text{cm/s})$
0.5	20.6
1.0	28.5
1.5	35.5
2.0	42.0
2.5	46.3
3.0	50.5

Fig. 7.4

- (i) On Fig. 7.5, plot the graph of $v/(\text{cm/s})$ on the y -axis against d/cm on the x -axis. Start your graph at $v = 20\text{ cm/s}$ and $d = 0$. Draw a smooth curve of best fit.

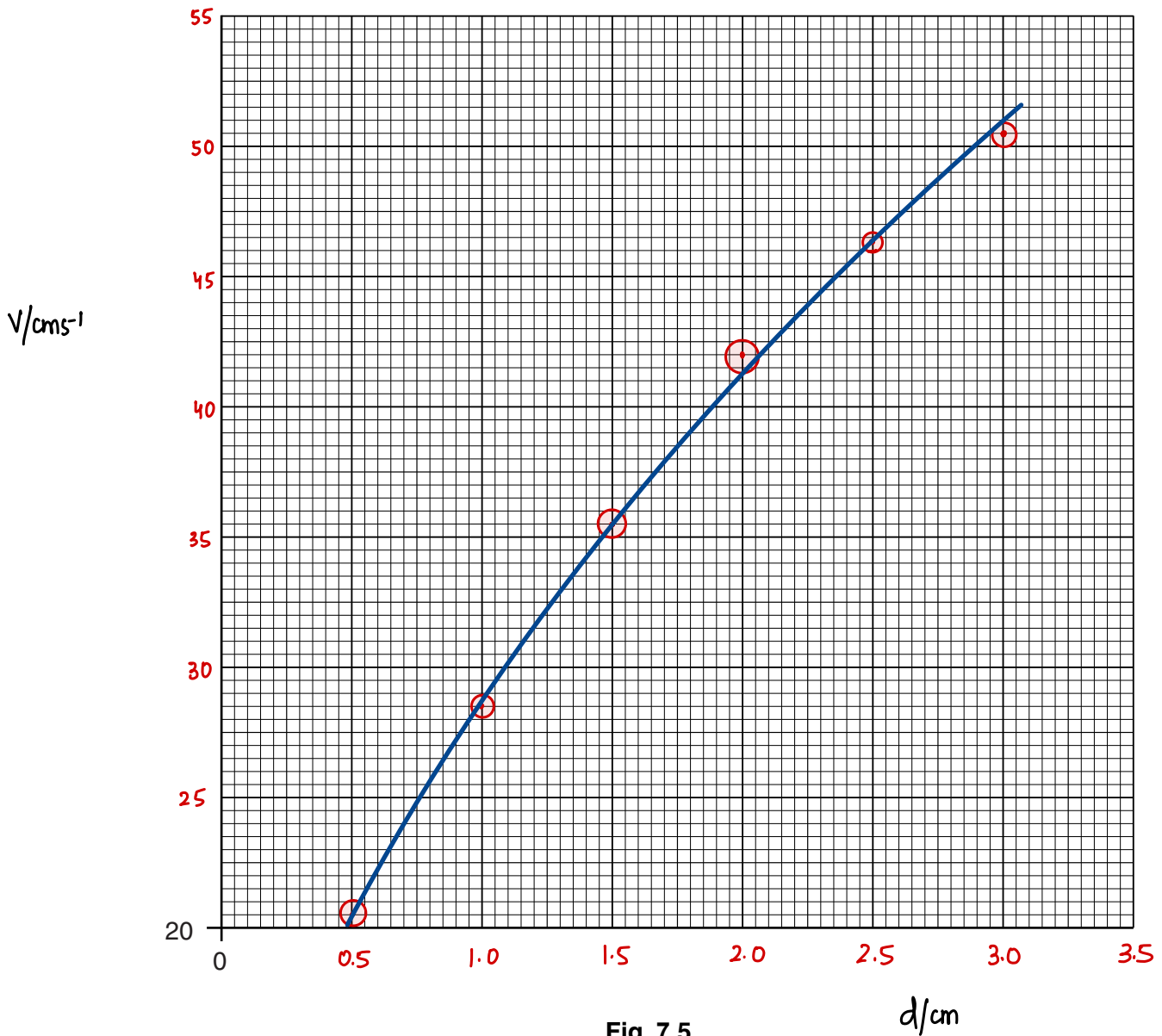


Fig. 7.5

- (ii) Suggest a possible reason why the student cannot take readings for d greater than 3.0 cm.

More than 3cm depth is not possible as tray is shallow, water will spill out.

..... [1]

- (iii) Using your graph, suggest how the speed varies with depth for very large values of d .

Speed of the wave will decrease as the depth increases & will become constant.

..... [1]

[ON2012/P42/Q1]

- 8 A student investigates the deviation of a ray of light by a prism.

A ray of red light passes through the prism. The student places pins P_1 and P_2 on the incident ray and pins P_3 and P_4 on the emergent ray.

Fig. 8.1 on page 11 shows the positions of the pins and the prism.

- (a) (i) On Fig. 8.1, draw a straight line through P_1 and P_2 .
Continue this line through the prism. [1]

- (ii) On Fig. 8.1, draw a straight line through P_3 and P_4 .
Continue this line to cross the line drawn in (i). [1]

- (iii) Measure the smaller angle between the line drawn in (i) and the line drawn in (ii).

angle = 36° [1]

- (iv) On Fig. 8.1, draw a line to represent the actual path of the ray through the prism. [1]

- (b) Suggest why the student uses red light and not white light in this experiment.

Prism will disperse light & seven colours will cause confusion.

..... [1]

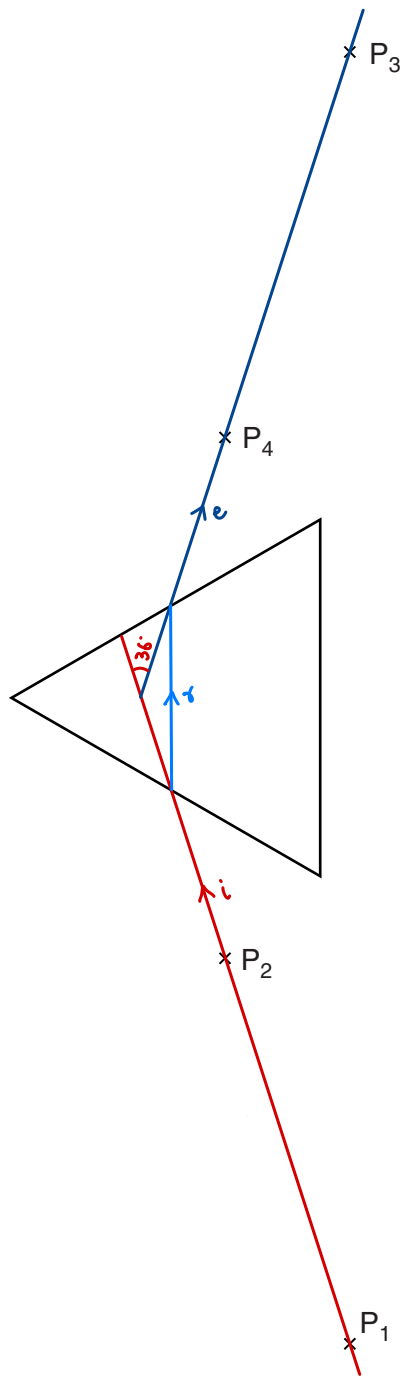


Fig. 8.1

- (c) In a different experiment, the student is asked to draw the normal at the point where an incident ray strikes a prism.

The student draws a line, as shown in Fig. 8.2.

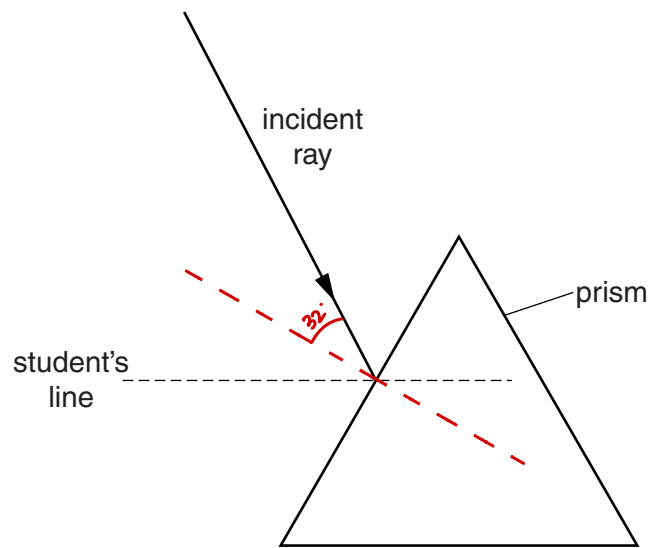


Fig. 8.2

- (i) Explain why the student's line is not the normal.

The normal is not making 90° with surface.

[1]

- (ii) On Fig. 8.2, draw the correct normal. Measure the angle of incidence i .

$i = 32^\circ$ [1]

[ON2012/P42/Q4]

- 9 A student determines the focal length of a lens.

The apparatus is set up as shown in Fig. 9.1.

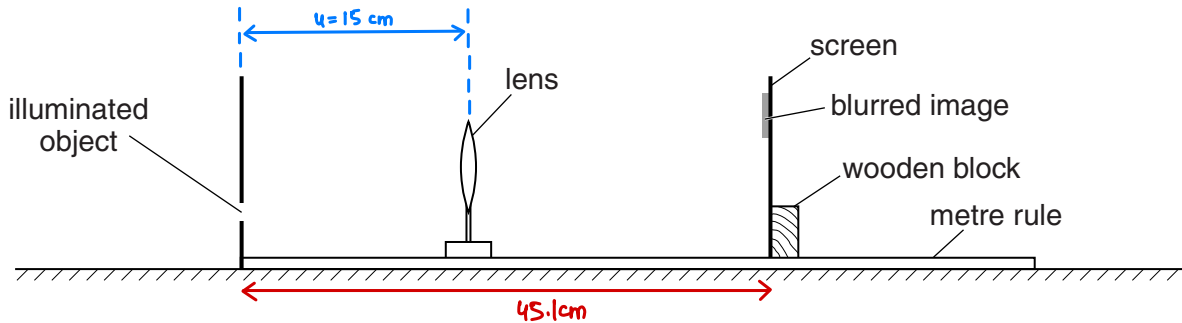


Fig. 9.1 (not to scale)

The illuminated object is fixed at the 0 cm mark on the metre rule.
The perpendicular distance u of the object from the lens is fixed at 15.0 cm.

- (a) On Fig. 9.1, mark u and label it 15.0 cm. [1]

- (b) The image on the screen is not clear and the centre of the image is above the centre of the lens.

Explain how the apparatus is adjusted

- (i) to produce a focussed image on the screen,

Move the lens away or towards the screen until sharp image is produced on screen. [1]

- (ii) so that the centre of the image is level with the centre of the lens.

Move the object upwards. [1]

- (c) The image on the screen is in focus.

The student measures the distance from the object to the screen.

Fig. 9.2 shows an enlarged view of part of the screen and the metre rule.

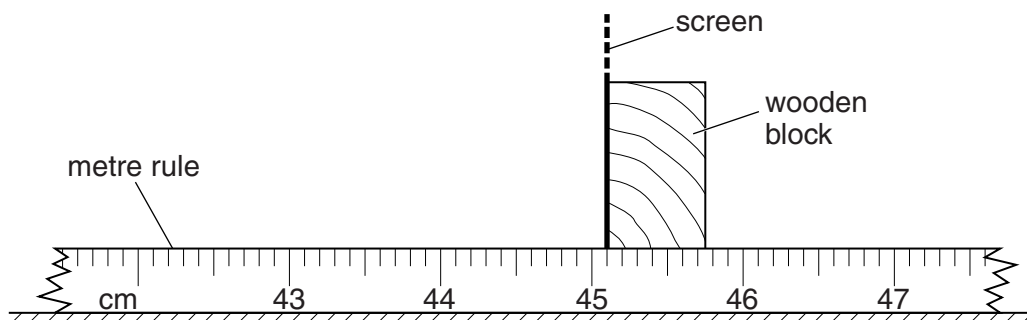


Fig. 9.2

- (i) State the reading on the metre rule at the position of the screen.

reading = 45.1 cm [1]

- (ii) Use your answer to (c)(i) to calculate the distance v of the image from the lens.

$$v = 45.1 - 15$$

$$v = 30.1 \text{ cm} \quad [1]$$

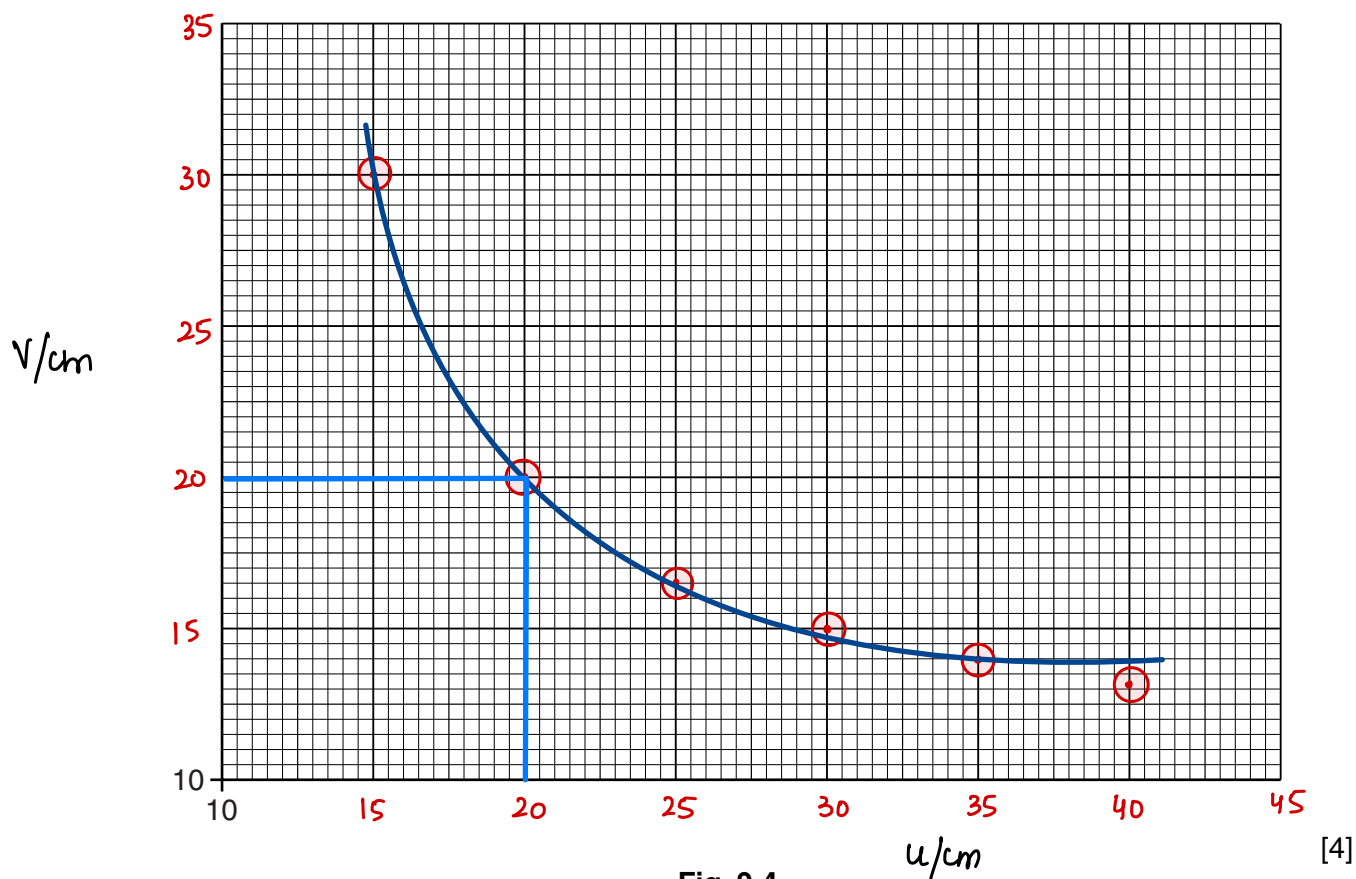
- (d) The student repeats the experiment for increasing values of u . The results are recorded in Fig. 9.3.

u/cm	v/cm
15.0	30.1
20.0	19.8
25.0	16.5
30.0	15.1
35.0	14.2
40.0	13.3

Fig. 9.3

- (i) In the spaces in Fig. 9.3, write your value of v from (c)(ii) and the corresponding value of u . [1]

- (ii) On Fig. 9.4, plot the graph of v/cm on the y -axis against u/cm on the x -axis. Start your graph from $v = 10 \text{ cm}$ and $u = 10 \text{ cm}$. Draw a curved line of best fit.



- (e) Suggest two practical techniques that the student uses to make the readings for v as accurate as possible.

1. Avoid parallax error by keeping an eye perpendicular to the reading.

.....

2. Place ruler close to the apparatus.

.....

[2]

- (f) When $u = v$, theory shows that the focal length f of the lens is given by $f = u/2$.

Use your graph to determine a value for f .

$$f = \frac{u}{2} = \frac{20}{2}$$

$$f = 10 \text{ cm} \quad [1]$$

[MJ2013/P42/Q1]

- 10** A student starts an experiment to determine the path of light through a glass block.

The student uses a rectangular glass block and optical pins. He places the glass block on a sheet of paper on a soft board, as shown in Fig. 10.1.

The student draws a line L to represent an incident ray.

- (a)** **(i)** On Fig. 10.1, draw the normal at the point where L meets the glass block. [1]

- (ii)** On Fig. 10.1, label the angle of incidence i . Measure i .

$i =$ [1]

- (b)** The student then places a pin P_1 on the incident ray, as shown in Fig. 10.1.

- (i)** Suggest a reason why the student uses a board under the piece of paper.

.....
[1]

- (ii)** The student places a second pin P_2 on L. Suggest where, on L, pin P_2 should be placed.

.....[1]

- (iii)** To find the path of the emergent ray, the student views P_1 and P_2 through the glass block.

He moves his head until P_1 and P_2 are in line.

Explain why it is important for the pins to be vertical.

.....
[1]

- (c)** The student finds it difficult to mark the emergent ray with the apparatus as it is set up in Fig. 10.1.

Suggest and explain a reason for this.

.....

[2]

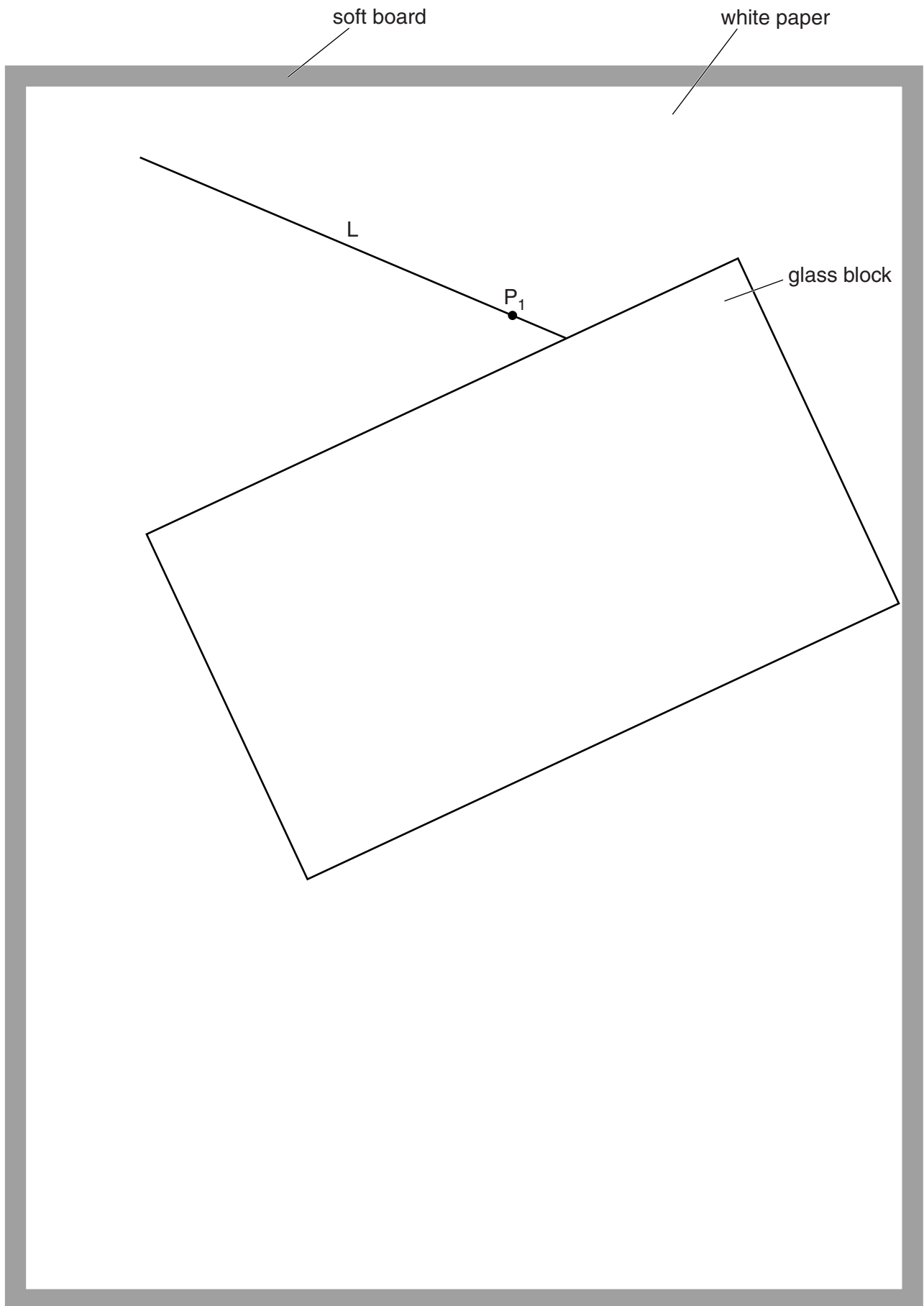


Fig. 10.1 (full scale)

[ON2013/P41/Q3]

11 A student investigates the reflection of light.

The student has available:

- a pin board,
- a sheet of plain paper,
- a plane mirror,
- optical pins,
- a pencil, ruler and protractor.

(a) Describe how the student uses the apparatus to verify that the angles of incidence and reflection are equal.
Include a clear labelled diagram in your answer.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) The student carries out the experiment carefully.
Describe one practical technique that improves the accuracy of the experiment.

.....

.....

..... [1]

12 A student is given a plano-convex lens.

Fig. 12.1 shows the plano-convex lens. It has one curved face and one flat face. The lens is a converging lens.

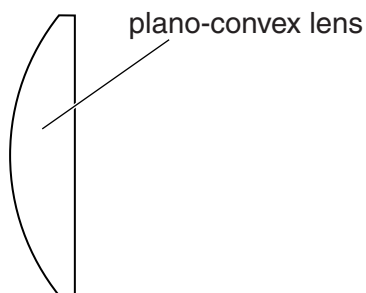


Fig. 12.1

The student predicts that the lens has two **different** focal lengths, one on each side of the lens.

Describe an experiment the student can perform to test this prediction.

In your account you should:

- state any equipment used,
- draw a labelled diagram of the equipment used,
- describe how the equipment is used,
- explain how the student checks his prediction.

[4]

- 13 An experiment is carried out to investigate refraction of light through a glass block.

Fig. 13.1 shows a rectangular glass block. A ray of light is incident at P at an angle of incidence of 40° . The angle of refraction in the block is 24° .

$\angle i$

$\angle r$

- (a) On Fig. 13.1, draw lines to represent

(i) the normal at P,

[1]

(ii) the refracted ray.

[1]

- (b) The lower face of the block is labelled XY.

(i) On Fig. 13.1, continue the normal to meet XY. Label this point A.

(ii) On Fig. 13.1, continue the line of the refracted ray to meet XY. Label this point B.

(iii) Measure AB and PB.

AB = 2.7cm

PB = 6.5cm

[1]

(iv) On Fig. 13.1, continue the line of the incident ray to meet XY. Label this point C.

(v) Measure AC and PC.

AC = 5.2cm

PC = 7.7cm

[1]

(vi) Theory suggests that the refractive index of the glass is given by the ratio

$$\frac{AC \times PB}{AB \times PC}$$

Calculate this ratio.

2SF

Give your answer to a suitable number of significant figures.

$$\frac{5.2 \times 6.5}{2.7 \times 7.7} = 1.62$$

ratio = 1.6 [1]

- (c) On Fig. 13.1, draw a line to represent the ray of light that emerges from the block.

Label this line L.

[1]

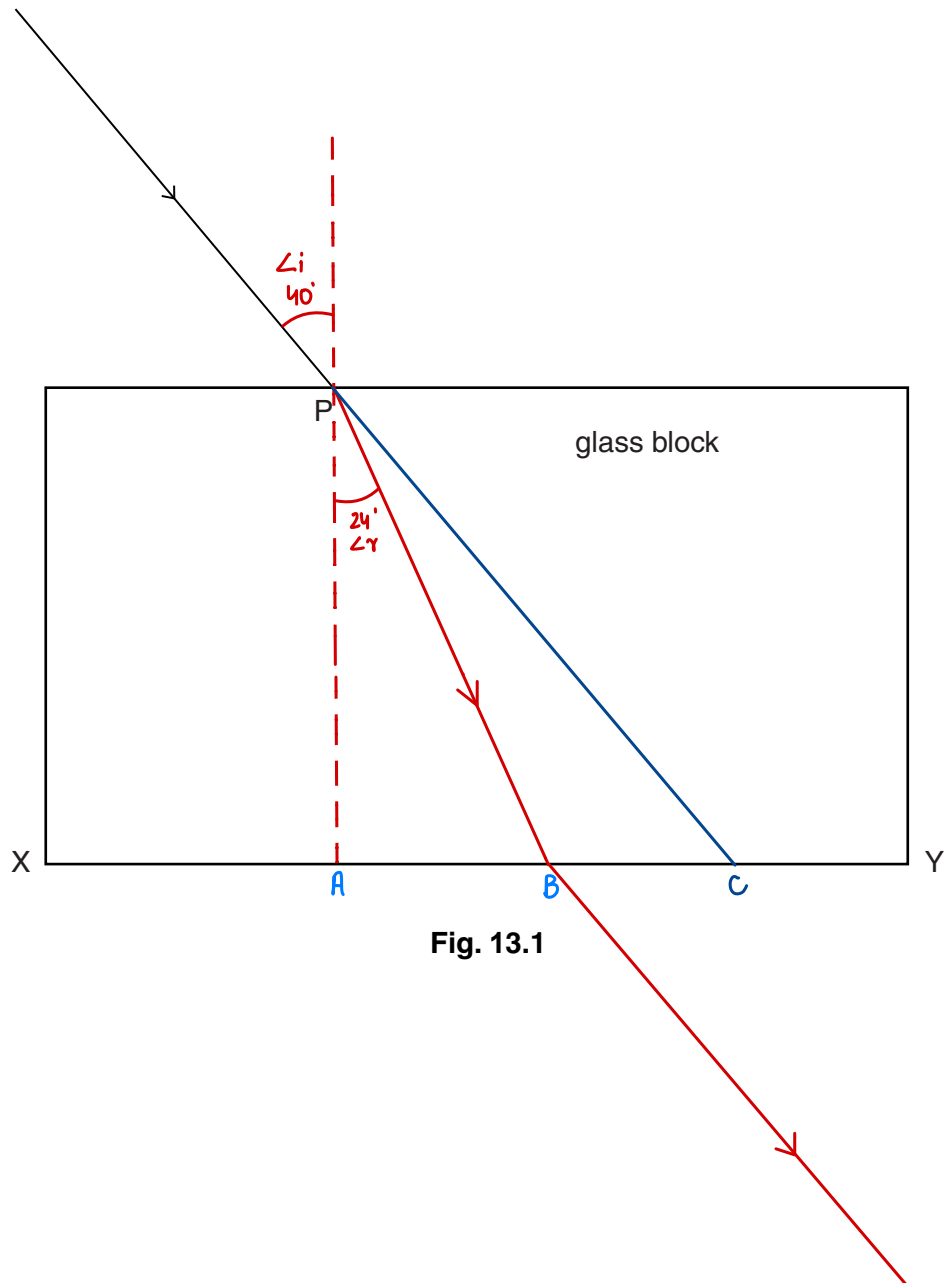


Fig. 13.1

- 14** A student determines the refractive index of the glass in a semi-circular block.

The student uses the relationship

$$\text{refractive index} = \frac{\sin(\text{angle of incidence})}{\sin(\text{angle of refraction})}$$

Describe an experiment the student can perform to obtain an accurate value for the refractive index of the glass. Normal laboratory equipment is available.

In your description of the experiment you should

- state the equipment used,
- describe how the equipment is used,
- state the readings taken,
- explain how the refractive index is obtained from the readings,
- describe how the student makes the experiment accurate.

The semi-circular glass block is drawn for you in Fig. 14.1. You may draw on Fig. 14.1.

[5]

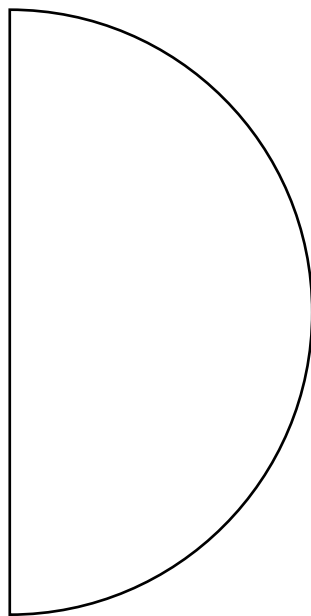


Fig. 14.1

- 15 A student uses a pin-hole camera to produce an image.

The pin-hole camera consists of a light-proof box with a small hole (pin-hole) on one side and a very thin paper screen on the opposite side, as shown in Fig. 15.1.

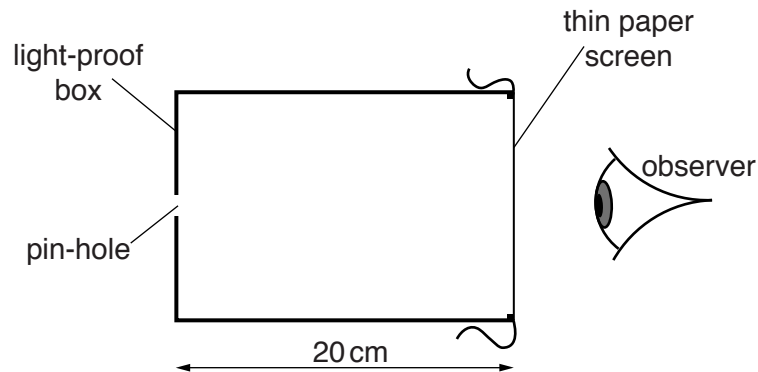


Fig. 15.1 (not to scale)

The student points the pin-hole of the camera towards a bright window, as shown in Fig. 3.2.

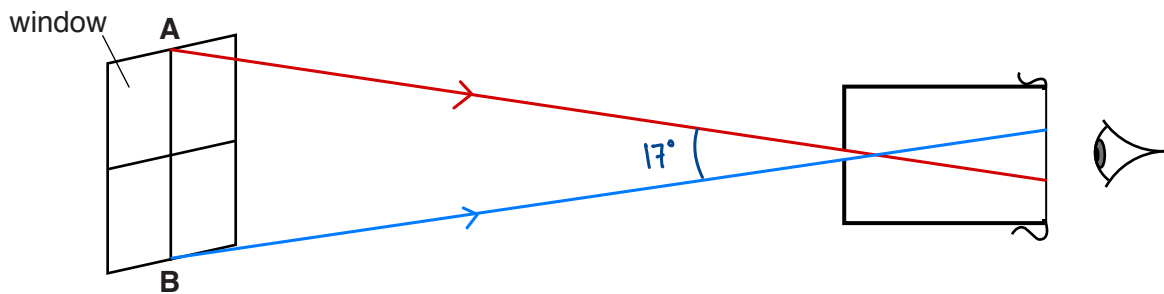


Fig. 15.2

Light enters the camera through the pin-hole and produces an image on the thin paper screen. The student can see the image through the thin paper. The student observes the image of the window on the screen.

(a) (i) On Fig. 15.2,

1. draw a ray of light from the top A of the window, through the pin-hole to the screen, [1]
2. draw a ray of light from the bottom B of the window, through the pin-hole to the screen, [1]
3. measure the angle between the two rays of light.

angle = 17° [1]

- (ii) Explain how you measured the angle as accurately as possible.

Measure angle from left as it has wider rays by avoid parallax error.

..... [1]

- (iii) Explain why the image of the window on the screen is inverted.

Image of window is real image which is always inverted

.....

..... [1]

- (b) The pin-hole allows only a small amount of light into the camera, so that the image of the window is not very bright. The student decides to make the pin-hole larger to allow more light into the camera, as shown in Fig. 15.3.

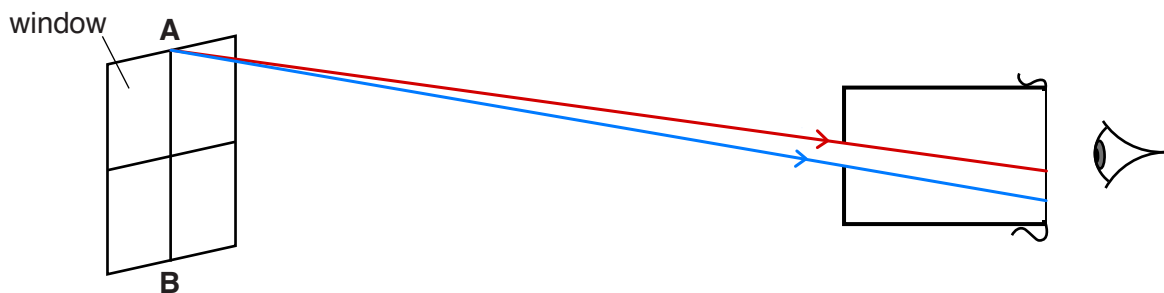


Fig. 15.3

The image formed on the paper screen is now blurred.
On Fig. 15.3, draw two rays from A that enter the camera.
Use your diagram to explain why the image is blurred.

Light from A has spread out at two different points so image is blurred.

..... [1]

[ON2015/P41/Q3]

- 16** A student performs an experiment to obtain an accurate value for the focal length of a converging lens.

His school has lenses with focal lengths 10 cm and 15 cm.

The student is given a lens from a packet labelled 'focal length 10 cm'.

- (a)** Describe a simple method the student can use in order to check that the lens has a focal length of 10 cm. You may use a diagram in your answer.

.....

 [1]

- (b)** The student then uses the apparatus in Fig. 16.1 to obtain an accurate value for the focal length f of the lens.

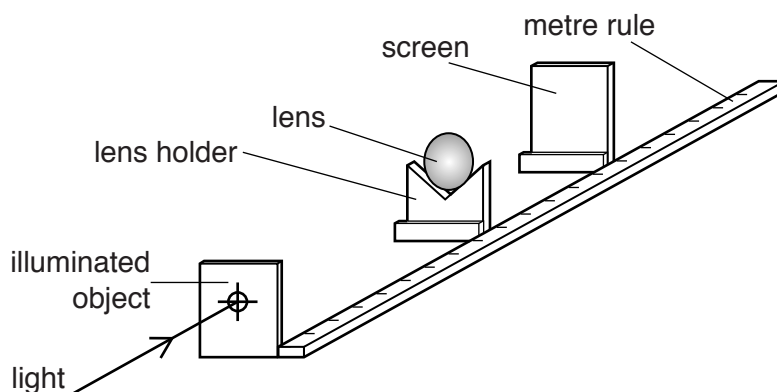


Fig. 16.1 (not to scale)

The student places the lens a measured distance u from the illuminated object. He then adjusts the position of the screen until a clear focused image is seen on the screen. He measures the distance D from the object to the focused image on the screen.

- (i)** On Fig. 16.1, mark and label the lengths u and D . [2]
(ii) The distance u is set at 85.0 cm and the student measures the distance D . He repeats the experiment and obtains the following values, in cm, for D .

96.5 96.3 96.2 96.1 96.2

Calculate D_{av} , the average value of D .
 Give your answer to three significant figures.

$D_{av} =$ [1]

- (iii)** State one way in which the student can ensure that each measurement of D is accurate.

..... [1]

- (c) The student repeats the experiment for a range of values of u and obtains a value for D_{av} each time. The results are recorded in Fig. 16.2.

u/cm	D_{av}/cm
85.0	
70.0	81.0
50.0	62.3
25.0	41.6
18.0	40.5
15.0	45.1
12.0	69.5

Fig. 16.2

- (i) On Fig. 16.2, add your value of D_{av} for $u = 85.0\text{ cm}$ from (b)(ii).

- (ii) On Fig. 16.3, plot the graph of D_{av}/cm on the y -axis against u/cm on the x -axis. Start your axes from (0, 30).

The graph shows that D_{av} has a minimum value.
Draw the smooth curve of best fit.

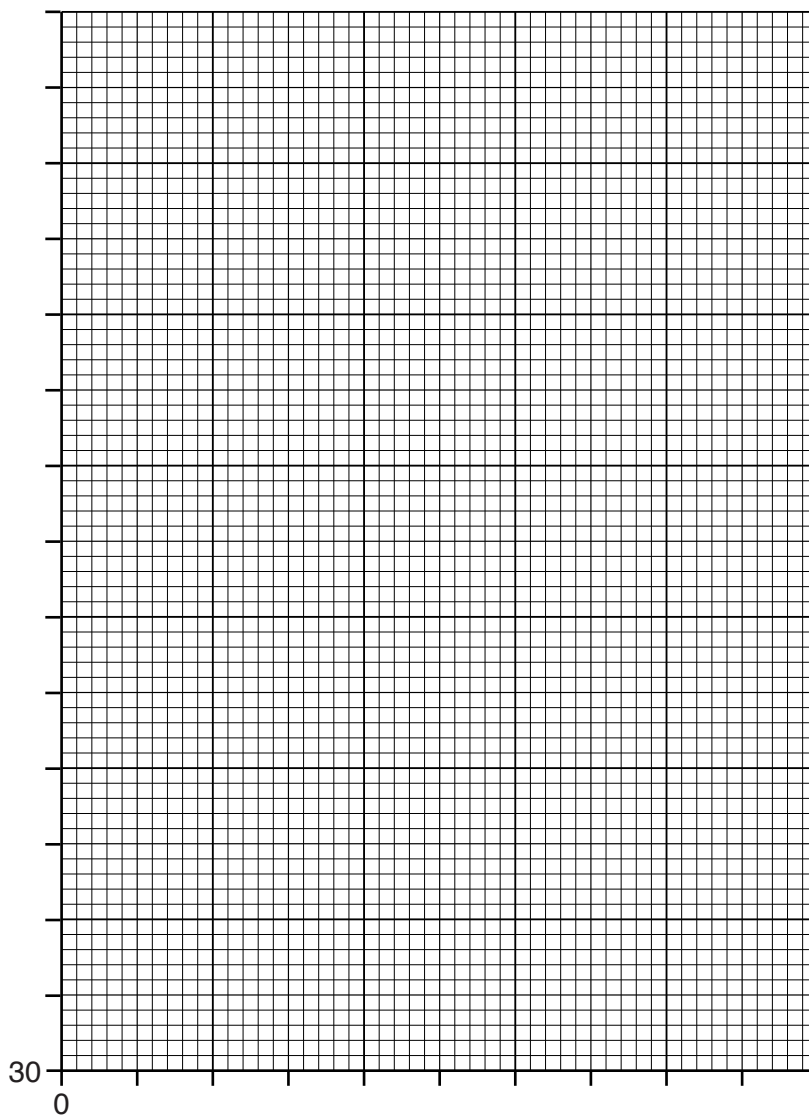


Fig. 16.3

[4]

- (iii) Use your graph to find

1. the minimum value of D_{av} ,

minimum value of $D_{av} = \dots\dots\dots$ [1]

2. u_m , the value of u when D_{av} is minimum.

$u_m = \dots\dots\dots$ [1]

- (iv) Theory shows that the minimum value for D_{av} is when $D_{av} = 4f$ and when $u_m = 2f$.

Calculate $\frac{D_{av}}{4}$ and $\frac{u_m}{2}$ from the values you have given in (c)(iii). Comment on your answers.

.....

..... [1]

[ON2015/P42/Q1]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – May/June 2011	5054	41

- 4 (a) (i) straight line from lamp to bench just above/touching top of card B1 [1]
(ii) correct indication of region of shadow B1 [1]
(b) shadow becomes longer B1 [1]
(c) multiple sources e.g.:
moon out
reflections
more street lamps
lights from other sources such as cars/houses
larger lamp size B1 [1]
[Total: 4]

Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
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- 2 (a) (i) accurate horizontal distance marked from **centre** of lens to screen B1 [1]
(ii) focal length / image distance B1 [1]
(b) repeat and average (measuring distance) B1
any TWO good practical points (may be marked on diagram) e.g.:
• adjust screen/lens distance to give clear image
• lens in holder
• lens and screen perpendicular to ruler / correct use of set square explained
• avoid parallax error in reading ruler/measuring f
• lens/screen close to ruler
• experiment in darkened room B2 [3]
allow alternative experiments to measure f **[Total: 5]**

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
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- 4 (a) (i) normal drawn correctly B1 [1]
(ii) i marked (ecf incorrectly drawn normal) B1 [1]
(iii) 29° to 31° (ecf (ii)) B1 [1]
(b) (i) line drawn parallel to and $5\text{ cm} \pm 1\text{ mm}$ below upper surface of block B1 [1]
(ii) line through (within 1 mm of) crosses M_1 and M_2 to candidate's line in (b)(i) B1 [1]
(iii) ray drawn within block B1 [1]
(c) 14° to 20° B1 [1]
[Total: 7]

Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
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- 1 (a) (i) move lens (along the ruler) / moves object and screen together (lens not moved) B1 [1]
(ii) (horizontally) align (centres) of object, lens and screen / raise object / lower lens B1 [1]
(allow raise screen)
(iii) any sensible answer for finding middle of side of block, e.g. how non-parallax used such as viewed from above measuring of length of block and divide by 2 B1 [1]
- (b) (i) 0.14 m cao B1 [1]
(ii) 0.245(1) m allow 0.25 m B1 [1]
- (c) (i) axes: labels correct way round, labelled quantity and unit B1
scales: more than $\frac{1}{2}$ grid, sensible, values consistent with labels B1
2 cm $\equiv$ 0.1 cm on both axes
points plotted accurately B1
straight line of best fit neatly drawn through all points B1 [4]
(ii) 0.97 to 1 ignore unit B1
correct use of at least half graph line ($\Delta D \geq 0.2$) shown on graph or in calculation B1 [2]
(iii) 0.24 m to 0.25 m B1 [1]
- (d) (more accurate because) gradient / more readings gives **average** (of different readings) / can ignore anomalous points / straight line from many/several points B1 [1]

[Total: 13]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
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- 4 (a) (i) ray drawn through M_1 and M_2 to block B1 [1]
(ii) ray completed accurately through block B1 [1]
(iii) $28^\circ \pm 2^\circ$ B1 [1]
- (b) increases angle of incidence described B1
until emergent ray along surface (described or drawn) B1 [2]
- (c) ray not along normal/not to middle of diameter/refracted on entry B1 [1]

[Total: 6]

Page 3	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – October/November 2012	5054	41

- 4 (a) to determine height accurately / to stop as soon as shoe moves B1 [1]
(b) $22^\circ \pm 1^\circ$ B1 [1]
(c) (i) any one sensible suggestion, e.g.
protractor has edge
protractor is small
divisions close together
alignment of zero difficult
board sags
board may move B1 [1]
(ii) **measures** two sides of triangle and uses trig formula B1 [1]
(may be shown on diagram)
(d) (better grip) larger angle / ramp lifted higher or reverse argument B1 [1]

[Total: 5]

Page 2	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – October/November 2012	5054	42

- 1 (a) time (for one length) is small/
reaction error large (compared with time)/
gives average B1 [1]
- (b) (i) distance (travelled by wave) / length of tray varies/
cannot place rule close to water B1 [1]
(ii) clear sensible description, e.g.
ruler viewed from vertically above / perpendicularly
use tape measure (flexible)
answers may be on diagram, e.g.
ruler and eye above
ruler and set squares B1 [1]
- (c) (i) zero error / dead space at end of ruler/
zero on ruler not at end/base of tray B1 [1]
(ii) use second ruler to measure length of dead space B1
length of dead space added to ruler reading (of depth) B1 [2]
- (d) (i) axes: correct way round, labelled quantity and unit
scales: more than $\frac{1}{2}$ grid, linear, not awkward
e.g. x-axis: 2 cm $\equiv$ 0.5 cm y-axis: 2 cm $\equiv$ 5 cm/s B1
points plotted accurately within $\frac{1}{2}$ small square
neat crosses or small points (in circle) B1
smooth curve of best fit drawn B1 [4]
- (ii) tray not deep enough/
water spills out (when wave made)/
time too small / wave travels too fast B1 [1]
- (iii) speed becomes constant / does not vary (with d)/
increase in speed becomes smaller (with increasing d)/
gradient decreases (with increasing d) B1 [1]
- [Total: 12]

Page 4	Mark Scheme	Syllabus	Paper
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- 4 (a) (i) line correctly drawn through P_1 and P_2 and extended into prism B1 [1]
(ii) line correctly drawn through P_3 and P_4 and extended back to cross (a)(i) B1 [1]
(iii) correct construction lines and $36^\circ \pm 2^\circ$ B1 [1]
(iv) path through prism correctly drawn B1 [1]
- (b) spectrum formed/
dispersion occurs/
splits into colours B1 [1]
- (c) (i) not perpendicular/at 90° to surface/prism/side of prism
normal is perpendicular to surface B1 [1]
(ii) correct normal seen and $32^\circ \pm 2^\circ$ B1 [1]
- [Total: 7]

Page 2	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – May/June 2013	5054	42

- 1 (a) accurate horizontal line from object to centre of lens
and labelled u or 15 cm B1 [1]
- (b) (i) move screen (along ruler) B1 [1]
(ii) raise object B1 [1]
- (c) (i) 45.1 cm cao unit required B1 [1]
(ii) 30.1 cm ecf (c)(i) – 15.0 B1 [1]
- (d) (i) 15.0 and (c)(ii) inserted into top line of table B1 [1]
(ii) axes: correct way round, labelled quantity and unit
scales: more than $\frac{1}{2}$ grid, linear, not awkward
y-axis e.g.: 2 cm $\equiv$ 5 cm x-axis e.g.: 2 cm $\equiv$ 5 cm B1
points plotted accurately within $\frac{1}{2}$ small square
neat crosses or small points (in circle) B1
smooth curve of best fit drawn B1 [4]
- (e) any two from:
repeat (the measurement of v) and average
avoid parallax in **reading** ruler **or**
eye line/line of sight perpendicular to scale/reading **or**
lens or screen close to ruler **or**
mark centre of lens on base of holder
use of set-square described
check for zero error on ruler
use darkened room
clear explanation of focussing
e.g. move screen from left, then from right
move through focussed image from both directions, then stop B2 [2]
- (f) 9.8 to 10.0 cm ecf graph unit required B1 [1]

[Total: 13]

Page 3	Mark Scheme	Syllabus	Paper
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- 3 (a) (i) correct normal drawn B1 [1]
(ii) $42^\circ \pm 1^\circ$ B1 [1]
- (b) (i) so pins stand up
so not to damage bench B1 [1]
(ii) pins far apart
pins more than 5 cm apart
P2 near left hand edge of board/at end of L
P2 near edge of paper B1 [1]
(iii) to be above line
cannot line up pins otherwise or
can only use base of pins B1 [1]
- (c) pins on emergent ray mentioned B1
pins too close
will not fit on paper
block too near edge
emergent ray too short B1 [2]

Page 2	Mark Scheme	Syllabus	Paper
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- 2 (a) diagram showing paper and plain mirror
plus incident and reflected rays **OR** four roughly correct pins B1
2 pins placed on incident ray B1
pins or image (of pins) viewed in/through mirror B1
lines drawn **and** angles i and r **measured** to normal B1 [4]
- (b) sensible suggestion, e.g.
view bottom of pins
pins vertical
pins far apart, e.g. greater than 5 cm
repeat for different angles/repeat experiment
sharp pencil B1 [1]

Page 4	Mark Scheme	Syllabus	Paper
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- 4 use of object and screen **or** use of ray box and paper (with cylindrical lens) [B1]
 how image focused on screen described
 e.g. distance from lens to screen varied
 e.g. two rays crossing on paper [B1]
 lens reversed [B1]
 correct reference to prediction
 e.g. no longer in focus [B1]
[4]

Page 3	Mark Scheme	Syllabus	Paper
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- 3 (a) (i) normal correct at P [B1]
 (ii) angle r correct $\pm 1^\circ$ [B1]
 (b) (iii) 2.8 ± 0.1 cm [B1]
 6.9 ± 0.1 cm unit required on at least **one** response
 (v) 5.3 ± 0.1 (cm) [B1]
 8.2 ± 0.1 (cm)
 (vi) 1.6 or ecf correct ratio calculated no unit [B1]
 (c) emergent ray drawn parallel to incident ray and **labelled L** [B1]
[6]

Page 3	Mark Scheme	Syllabus	Paper
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- 3 a correct experiment described [M0]
 i.e. must be refraction
 ray box OR pins AND [B1]
 protractor AND ruler AND
 any one from
 (plain) paper / board / (sharp) pencil
 mark ray in air on both sides of block with pins or crosses [B1]
written description of: [B1]
 join points in air to block (both sides) and
 (remove block to) draw ray in block
 correct angles measured and labelled on diagram [B1]
 or described if no diagram drawn
 accuracy mark: e.g. [B1]
 repeats described anywhere
 fine pencil
 pins far apart
 bottom of pins
 large angles
 vary angle of incidence

[Total marks: 5]

Page 3	Mark Scheme	Syllabus	Paper
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- 3 (a) (i) ray from A to screen through pinhole B1
 ray from B to screen through pinhole B1
 $17 \pm 2^\circ$ B1
 (ii) use of longer lines, e.g. use of rays on left hand side of pinhole/extend rays
 on right/measure the angle from both rays/measure angle at both sides of
 the inter section B1
 (iii) ray from A forms bottom of image ora with B/ image is real /rays of light
 cross or intersect/ rays of light meet at the pinhole (before the screen) B1
 (b) light from A spread out/diverge on screen/form more than one image/ rays from
 A do not meet (at a point on screen) B1

[Total: 6]

Page 2	Mark Scheme	Syllabus	Paper
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1	(a) use of distant object <u>and</u> screen		B1
	(b) (i) u marked correctly		B1
	D marked correctly		B1
	(ii) 96.3 cm unit required		B1
	(iii) avoid parallax in <u>reading</u> (ruler)/view perpendicularly		
	lens/screen close to ruler		
	use darkened room		
	check for zero error		
	clamp ruler		
	move screen (not lens) backwards and forwards (until sharp image found)		
	object and lens same height (above bench)		
	object and lens and screen perpendicular to bench/vertical		B1
	(c) (ii) axes correct way round, labelled quantity and unit		B1
	scales linear, sensible		B1
	points plotted accurately within $\frac{1}{2}$ small square		
	neat crosses or small points (in circle)		B1
	best fit smooth curve drawn		B1
	(iii) 1. 39 to 41 (cm)		B1
	2. 20 ± 2 (cm)		B1
	(iv) both values for f <u>correctly</u> calculated and sensible comment		B1
			[Total: 12]